An Introduction to Statistical Learning

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1. What is Statistical Learning?
2. Why Do We Estimate f?
3. How Do We Estimate f?
4. Prediction Accuracy vs. Model Interpretability
5. Supervised vs. Unsupervised Learning

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What is Statistical Learning?

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Basic assumptions for Statistical learning

1. Suppose we observe a set of observations $(Y_i, X_i), i = 1, 2, ..., n$.
2. $X_i = (X_{i1}, X_{i2}, ..., X_{ip})$ is a p dimensional vector.
3. We believe that there is a relationship between Y and at least one of the X, which is given by $Y_i = f(X_i) + \varepsilon_i$
4. f is an unknown function and ε is a random error with mean 0.

A Simple Example

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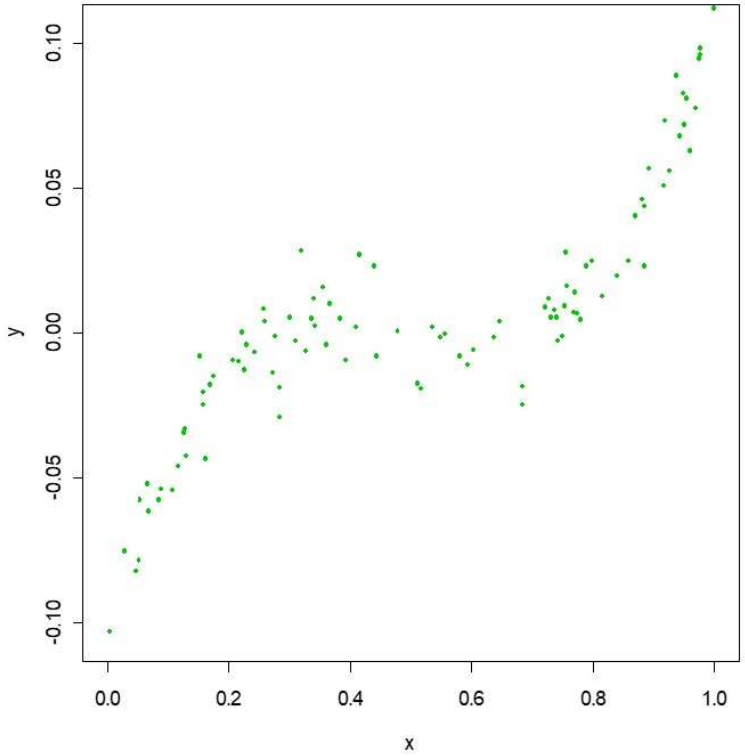

Figure 1: A Simulated data set

A Simple Example

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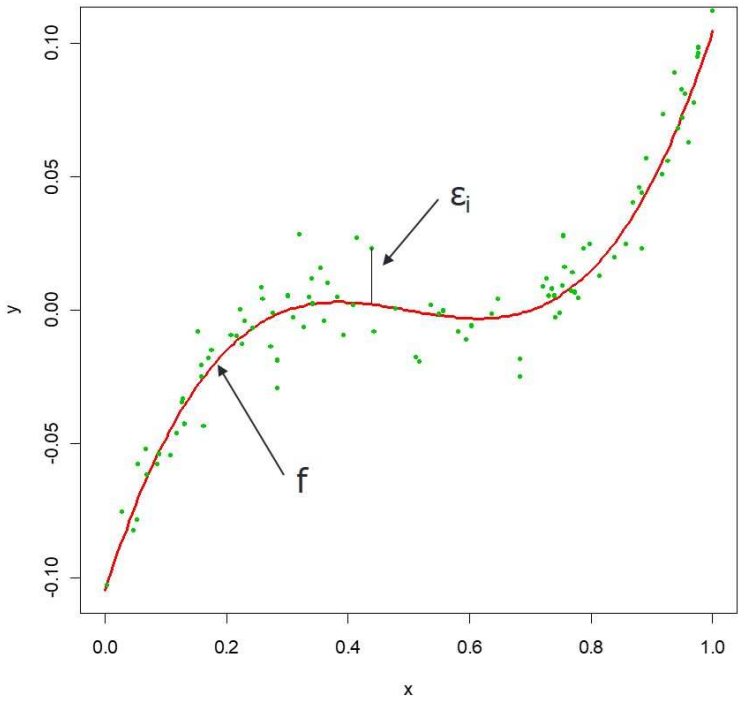

Figure 2: A Simulated data set with function f

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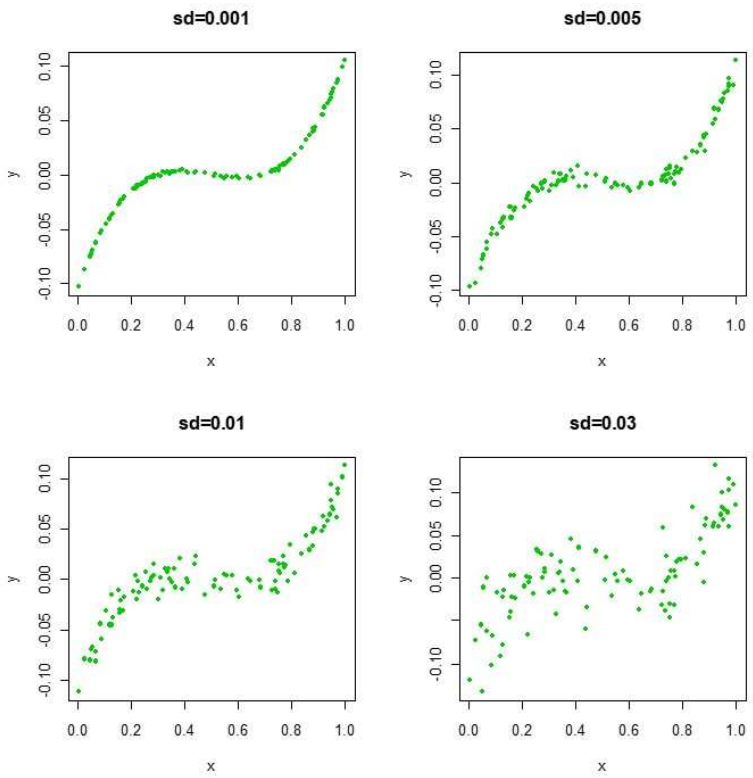

Figure 3: Simulated data sets with different ε

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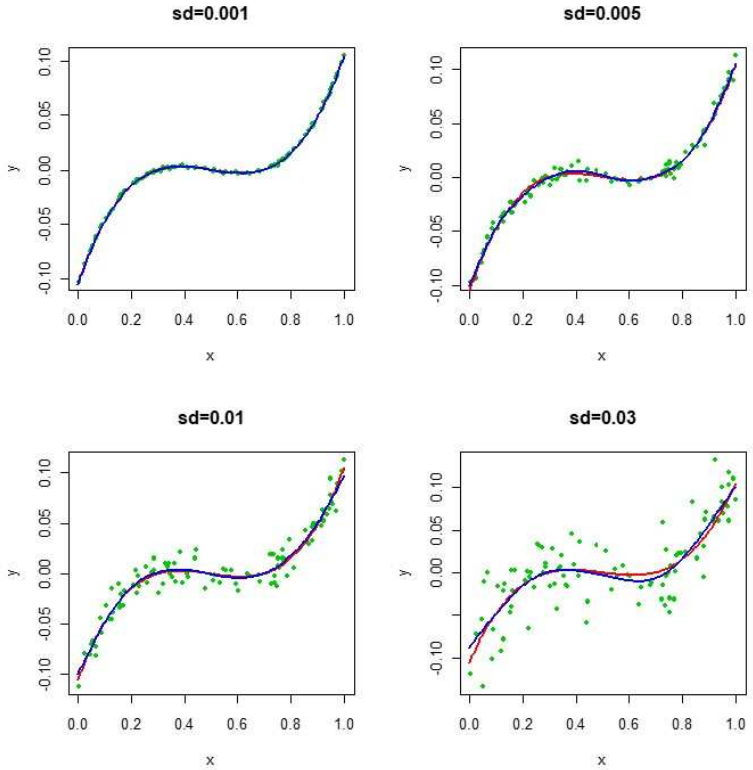

Figure 4: Simulated data sets with different estimates on function f

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Why Do We Estimate f?

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Facts we have to accept

1. We are trying to discover the true function f which describe the relationship between X (independent variables)and Y (dependent variable).
2. The true relationship f is difficult to find.

Two purposes for estimating f

1. Prediction.
2. Inference.

Prediction

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Given a good estimate for function f, we can make accurate predictions for the response, Y, based on a new value of X.

Example: Direct Mailing Prediction

1. Interested in predicting how much money an individual will donate.
2. Observations from 90,000 people with over 400 different characteristics are collected.
3. Don't care too much about each individual characteristic.
4. Just want to know: For a given individual should I send out a mailing?

Inference

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Alternatively, we may also be interested in the type of relationship between Y and the X.

Example: Housing Inference

1. Model the relationship between median house price and other 14 variables.
2. Which particular predictors actually affect the response?
3. Is the relationship positive or negative?
4. How much impact does one variable have on the house value.

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Parametric Methods

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It reduces the problem of estimating f down to one of estimating a set of parameters.

A two-step model for estimating f.

1. Make some assumption about the functional form of f and come up with a model. The most common example is a linear model.

$$f(X_i) = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_p X_{ip} \qquad (3.1)$$

2. Use the training data to fit the model or equivalently estimate the unknown parameters $\beta_j, j = 0, 1, 2, ..., p$.

Example: A Linear Regression Estimate

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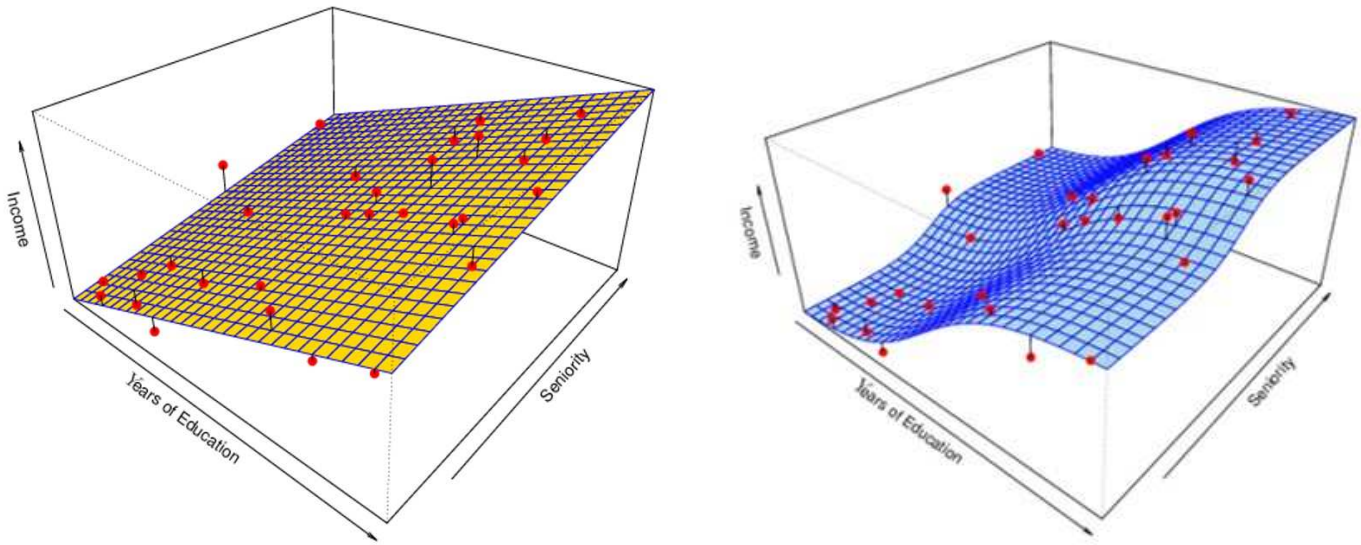
Even if the standard deviation is low we will still get a bad answer if we use the wrong model.



(a) Estimated (b) True

Figure 5: A Linear Regression with $f = \beta_0 + \beta_1 \times Eduation + \beta_2 \times Seniority$

Non-parametric Methods

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Non-parametric Methods do not make explicit assumptions about the functional form of f.

Advantages VS Disadvantages

1. Advantages: They accurately fit a wider range of possible shapes of f.
2. Disadvantages: A very large number of observations is required to obtain an accurate estimate of f.

Example: A Non-linear Regression Estimate

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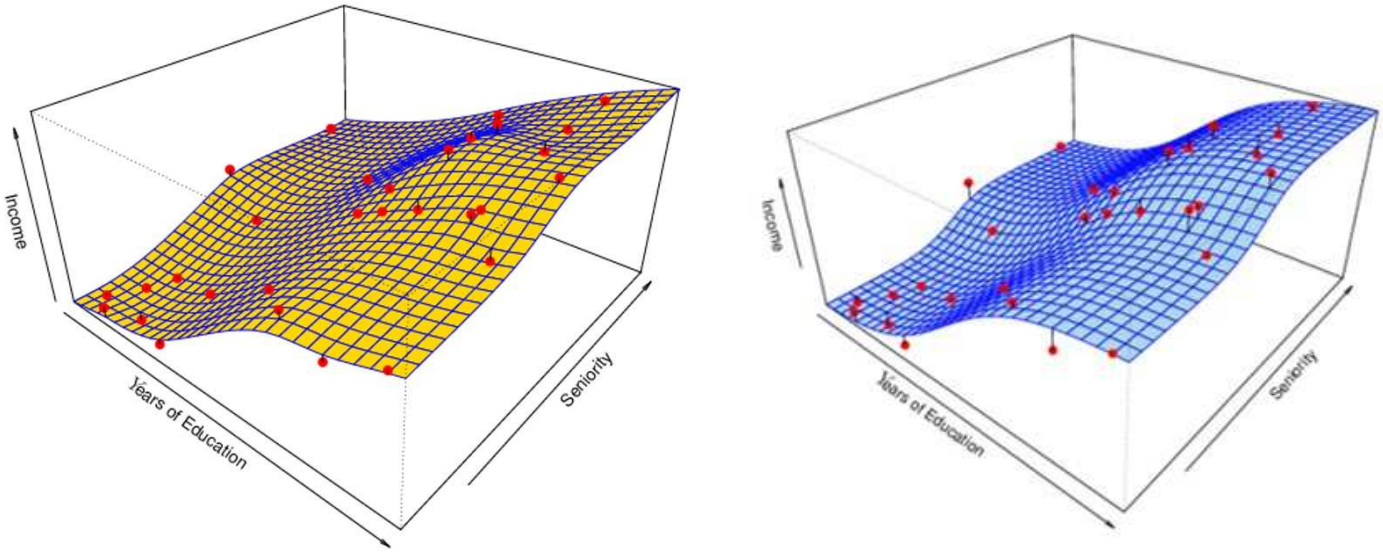
Non-linear regression methods are more flexible and can potentially provide more accurate estimates.



(a) Estimated (b) True

Figure 6: A Non-linear Regression Estimate

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Prediction Accuracy vs. Model Interpretability

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Why not just use a more flexible method if it is more realistic?

Two Major Reasons

1. A simple method produces a model which is much easier to interpret. For example, in a linear model, β_j is the average increase in Y for a one unit increase in X_j holding all other variables constant.
2. Even if you are only interested in prediction, it is often possible to get more accurate predictions with a simple, instead of a complicated model. It is harder to fit a more flexible model with limited data.

Example: A Poor Estimate

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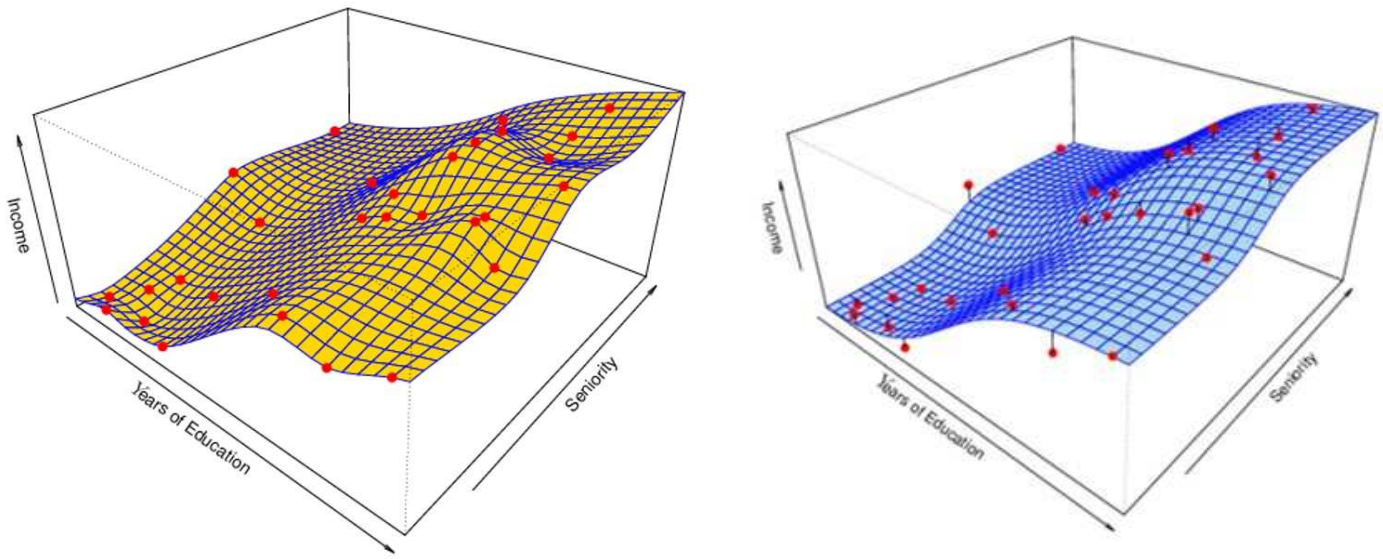
Non-linear regression methods can also be too flexible and produce poor estimates for f.



(a) Estimated (b) True

Figure 7: A Poor Non-linear Regression Estimate

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We can divide all learning problems into Supervised and Unsupervised situations

Supervised Learning

Supervised Learning is where both the predictors, X_i, and the response, Y_i, are observed.

Unsupervised Learning

1. Only X are observed.
2. Use X to guess what Y would have been and build a model from there.
3. A common example is market segmentation where we try to divide potential customers into groups based on their characteristics.

Regression vs. Classification

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Supervised learning problems can be further divided into regression and classification problems.

Regression

Regression covers situations where Y is continuous/numerical.

1. Predicting the price of a particular stock.
2. Predicting the value of a given house based on various inputs.

Classification

Classification covers situations where Y is categorical.

1. Is this email a SPAM or not?
2. Will the stock price increase or decrease?

A Simple Clustering Example

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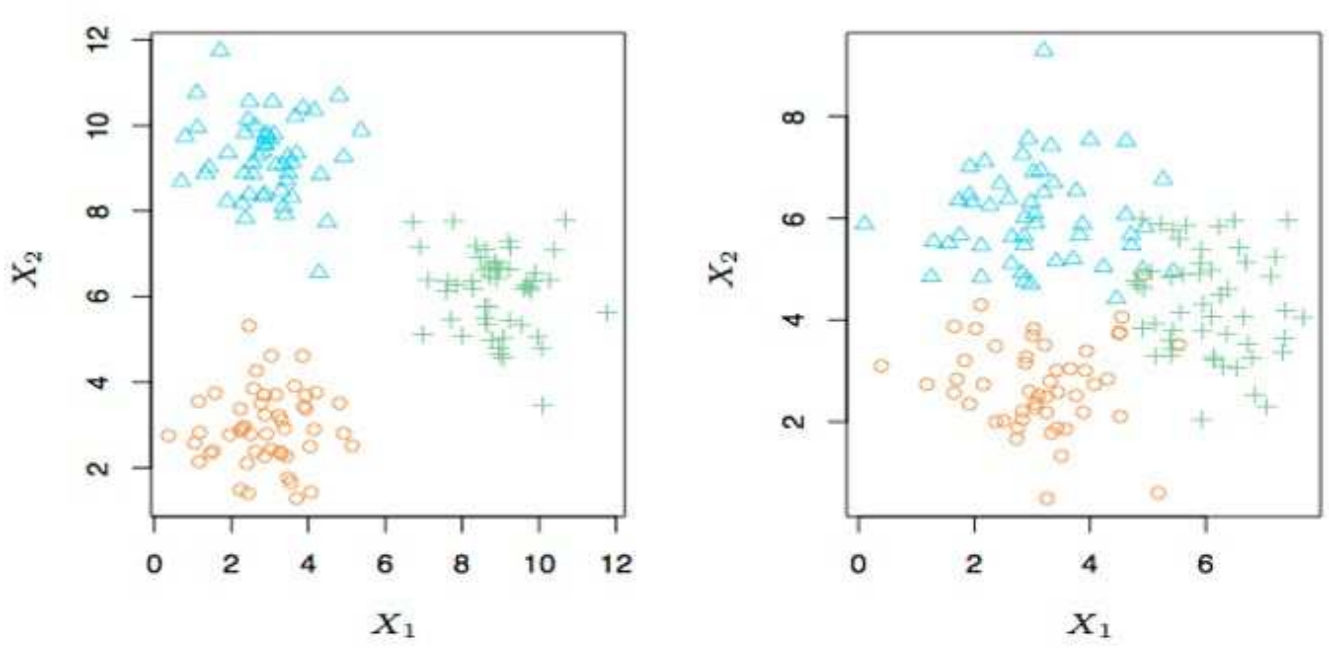

Figure 8: A Simple Clustering Example

Questions and Answers

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Thank you for your attention!